# Cambridge (CIE) A Level Computer Science

Your notes

## Translation Software

### Contents


- Translation Process
- Syntax Rules
- Expression Evaluation



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# Interpreter execution

## What is an interpreter?

- An interpreter is a type of translator that **executes high-level code line by line**
- It **does not** convert the entire program into a separate machine code file

## How does it work?

- The source code is **read, analysed, and executed one line at a time**
- No complete machine code file is produced, only **immediate execution** happens
- If there is an error in a line**, execution stops**, and the **error is reported immediately**

| Feature | Explanation |
|---|---|
| No output file | It does **not generate** a standalone .exe or machine code file |
| Immediate execution | Runs the code **as it is interpreted** |
| Slower execution | Each line is **translated every time** the program runs |
| Useful for development | Good for testing and debugging during coding |
| Stops on first error | Detects and reports errors **as they occur** |

## Example use cases

- Used in **educational tools**, like Python shells or BASIC interpreters
- Helpful for **rapid prototyping** or when frequent changes are made

# Compilation stages

## What is compilation?

- Compilation is a **process** that **translates a program** written in a high-level programming language into machine code
- **Only machine code** can be executed by a computer
- There are four stages involved in this process:
  - **Lexical Analysis**
  - **Syntax Analysis**


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- **Code Generation**
- **Optimisation**

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## Lexical analysis

- In A Level Computer Science, lexical analysis means studying the **words or vocabulary** of a language
- This stage involves **identifying lexical 'tokens'** in the code
- Tokens represent small meaningful units in the programming language, such as:
  - **Keywords**
    - var, const, function, for, while, if
  - **Identifiers**
    - Variable names, function names
  - **Operators**
    - '+', '++', '-', '*'
  - **Separators**
    - ', ' ';', '{', '}', '(', ')'
- During this stage, unnecessary elements like comments and whitespace are ignored
- For example, if the following code is being compiled:

```
var x = function(x,y) {

if(x>2) {

return x*y;

}

return x+y;
}
```

- The result of lexical analysis is a **token table**

| | Token | Type |
|---|---|---|
| 1 | var | Keyword |
| 2 | x | Identifier |
| 3 | = | Operator |
| 4 | function | Keyword |


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| | | |
|---|---|---|
| 5 | ( | Separator |
| 6 | x | Identifier |
| 7 | , | Separators |
| 8 | y | Identifier |
| 9 | ) | Separator |
| 10 | { | Separator |
| 11 | return | Keyword |
| 12 | x | Identifier |
| 13 | * | Operator |
| 14 | y | Identifier |
| 15 | ; | Separator |
| 16 | } | Separator |

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## Syntax analysis

- Now that tokens have been identified, syntax analysis makes sure they all **adhere to the syntax rules** of the programming language
- A symbol, e.g. '$' could be a valid token but not a valid character according to particular programming languages
- The dollar symbol would be flagged as breaking the syntax rules
- Other syntax errors programmers commonly make include **mismatched parentheses** or **missing semicolons**
- If the code passes the syntax analysis, the compiler can create an **Abstract Syntax Tree** (AST)
- An AST is a graph-based representation of the code being compiled
- An AST is an efficient way to represent the code for the next step

## Example abstract syntax tree

- For the same code as above, the following abstract syntax tree can be created


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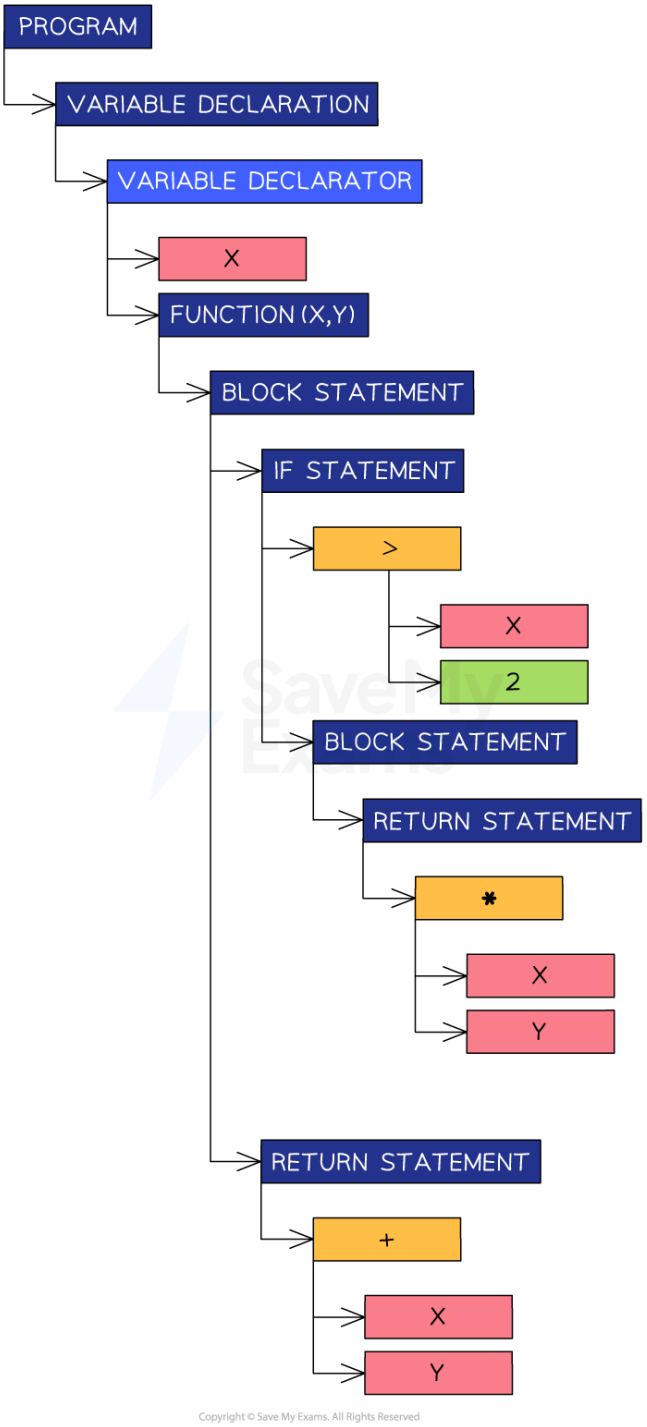


*Abstract syntax tree*

## Code generation

- This step takes the AST and traverses it to generate **object code** that can be executed by the computer

## Optimisation

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- This step **modifies the code** to make it **more efficient** without changing its functionality
- This is important to attempt because it **reduces the memory required** to run the code, which leads to faster execution
- A common optimisation action is **removing duplicate code**
- If an 'add' function is written twice in the source code, a sophisticated compiler will notice this and include it only once in the object code

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# BNF & syntax diagrams

- When writing programs, you must follow a set of strict rules known as **syntax** — the structure that defines how code must be written.
  - Just like spoken languages have grammar rules, **programming languages have syntax rules**
  - **Each language has its own syntax**, e.g. Python uses indentation, Java uses semicolons
- Programs are first written in **source code** (a high-level language like Python, Java, etc.)
- To run the program, this source code must be **translated into machine code**
- For translation to work, syntax must be **precisely defined** to avoid **ambiguity**
- Translators (compilers or interpreters) rely on these rules to correctly convert the program into a form the CPU can understand

## Regular expressions and meta-languages

- Regular expressions are used to **describe simple patterns** in text or syntax
- Many programming constructs (like variable names or simple statements) **can be defined** using regular expressions
- However
  - Some aspects of a programming language are **too complex** for regular expressions alone
- When regular expressions aren't enough, we use a **meta-language**
- A language used to **describe the structure of another language**
- The most common example is **Backus-Naur Form (BNF)** or **syntax diagrams**
- These allow us to define:
  - Full grammar rules
  - Nested structures
  - Hierarchical syntax

# What is Backus-Naur form (BNF)?

- BNF is a **meta-language**, a way of writing rules that define the **syntax of programming languages** or **data structures**
- BNF is used to:
  - Describe **formal grammar**


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- Define how statements and expressions are **structured**
- Ensure that syntax rules are **unambiguous** and **machine-readable**
- BNF uses a set of **rules** made up of:

| Term | Meaning |
|---|---|
| **Non-terminal** | A category or component (e.g. <digit>, <identifier>) |
| **Terminal** | An actual value or symbol (e.g. 0, 1, +, if) |
| **::=** | "Is defined as", separates the name from its definition |
| **\|** | "or", it gives alternative definitions |

## Example 1

<digit> ::= 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9
<number> ::= <digit> | <digit> <number>

- <digit> is defined as any single digit 0-9
- <number> is defined as:
  - a single digit, or
  - a digit followed by another number (i.e. multiple digits)

## Example 2

- <number> contains one or more digits
- BNF for <number> and use it to define <IDcode>, which consists of a <number> followed by either a <letter> or another <number>

<number> ::= <digit> | <number><digit>
<IDcode> ::= <number><letter> | <number><number>

- Recursive definition of one or more items
- Use of | for choice
- Application to a second rule with multiple components

# What is a syntax diagram?

- A syntax diagram is a **graphical representation of BNF** (meta-language)
- It uses **symbols** to map directly to the structure of BNF
- The symbols used are:

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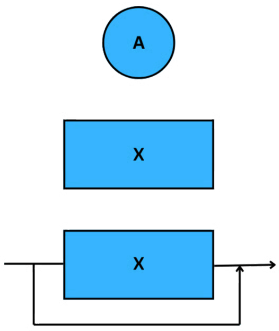


**Terminal element**
cannot be broken down any further

**Non terminal element**
will be broken down further in another diagram

**Non terminal element**
May be used more than once

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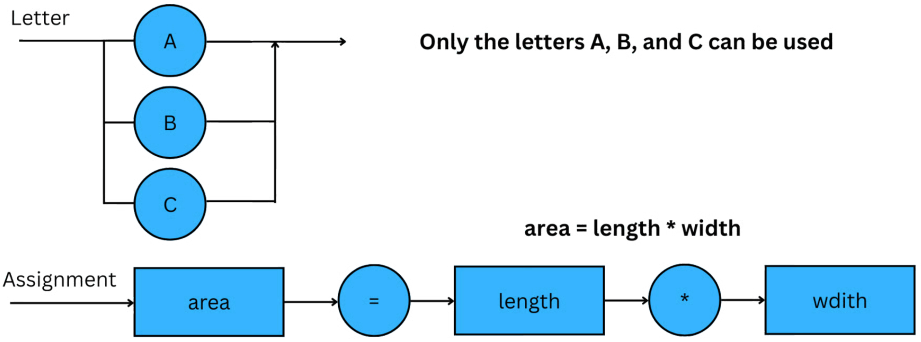
## Examples




**Only the letters A, B, and C can be used**

**area = length * width**

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# Reverse Polish Notation (RPN)

## What is RPN?

- Reverse Polish Notation (RPN) is a method of writing **mathematical expressions** where the **operator comes after the operands**
- This is also known as **postfix notation**
  - Infix: 3 + 4
  - RPN: 3 4 +

## Why use RPN?

- **No need for brackets**: The order of operations is determined by position, not parentheses
- **Easier for computers to evaluate** using a **stack**
- Always **unambiguous**

## How RPN works

- RPN uses a stack to evaluate expressions

1. **Read from left to right**
2. **Push numbers onto the stack**
3. When an **operator** is read:
   - **Pop the top two values** from the stack
   - **Apply the operator**
   - **Push the result** back onto the stack
4. When the expression ends, the final result is at the **top of the stack**

## Example: evaluate 3 4 + 2 ×

| Step | Action | Stack |
|---|---|---|
| Read 3 | Push to stack | 3 |
| Read 4 | Push to stack | 3 4 |
| Read + | Add $3 + 4 \rightarrow 7$ | 7 |
| Read 2 | Push to stack | 7 2 |


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| | | |
|---|---|---|
| Read × | Multiply 7 × 2 → 14 | 14 |

- Final result: 14

## Example: convert this infix expression to RPN

(7 − 2 + 8) / (9 − 5)

- **Split the expression into parts:**
- Left-hand side:

(7 − 2 + 8)

- This becomes:

( (7 − 2) + 8 )

- Which in RPN is:

7 2 − 8 +

- Right-hand side:

(9 − 5)

- In RPN:

9 5 −

- **Combine with the division**
- In RPN, that becomes:

7 2 − 8 + 9 5 − ÷

### Examiner Tips and Tricks

RPN **follows the evaluation order naturally**, so there's **no need for brackets** or precedence rules. Just:

- Work from inside brackets out
- Convert each sub-expression
- Add the operator after its operands

## Summary

| Concept | Description |
|---|---|
| RPN (Postfix) | Operators follow operands |

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| | |
|---|---|
| Stack | Used to store and evaluate values |
| Evaluation method | Push numbers, pop and evaluate on operator |
| Benefits | No brackets, no ambiguity, easy to compute with stack |

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